# MCP + RAG for 1C — one-page summary

Local-first search over BSL business logic and configuration metadata · Sergey Matyukhin

## PROBLEM

- A 1C configuration dump is thousands of .bsl modules plus a multi-megabyte metadata report; grep finds text, not meaning.
- An exact handler name and an intent question need different search strategies, and an LLM agent needs a tool with measured quality.
- Hosted search stacks require deployment work and send the code off the machine.

## APPROACH

- Local MCP server over stdio: 9 tools, including an explainable router and a diagnostic index_stats.
- intfloat/multilingual-e5-small (384-d, 512-token limit) + Chroma with HNSW cosine, CPU only.
- Custom global BM25 inverted index; Reciprocal Rank Fusion with k = 60; optional flashrank cross-encoder, off by default.
- No external APIs and no data leaving the machine; 40-60 ms per query.

## CORPUS

- 154,560 BSL code chunks (one procedure or function per chunk, ADR 0007) + 25,026 metadata chunks from the configuration XML = 179,586 chunks.
- Chunking gates: 99.6% code coverage; 0.3% of chunks over 512 tokens; token p50/p99/max = 179 / 472 / 554.
- Metadata source change (text report → XML): BM25 Hit@3 93.0% → 99.7% (ADR 0008).

## QUALITY — GOLD-SET V2: 70 NATURAL-LANGUAGE QUERIES, FILE-LEVEL RELEVANCE, 4 RUNS, MEAN ± STD

| Configuration | nDCG@10 | Recall@10 | Hit@1 | MRR@10 |
|---|---|---|---|---|
| vector | 0.339 ± 0.023 | 0.523 ± 0.030 | 0.150 ± 0.007 | 0.268 ± 0.019 |
| pearson | 0.337 ± 0.030 | 0.511 ± 0.048 | 0.179 ± 0.021 | 0.270 ± 0.024 |
| hybrid (RRF) | 0.301 ± 0.017 | 0.482 ± 0.034 | 0.129 ± 0.010 | 0.240 ± 0.013 |
| hybrid + rerank | 0.117 | 0.157 | 0.086 | 0.102 |

Paired test, vector vs hybrid: Δ nDCG@10 = +0.037, bootstrap 95% CI [-0.015; +0.086], p = 0.154 → NOT statistically significant. The default is unchanged; ADR 0009 is proposed.
2×2 ablation (nDCG@10, vector/hybrid): old export · symbolic 0.176/0.209; old export · procedural not measurable (dump gone); current export · symbolic 0.165/0.205; current export · procedural 0.339/0.301. Chunking explains the leader flip; the export effect at symbolic chunking is −0.012/−0.004, inside run noise. Interaction cell missing, additivity assumed. The cross-encoder stays an option only (0.117).

## ENGINEERING

- 72 unit tests; ruff clean, mypy over 22 modules, cyclomatic-complexity gate 143 blocks / avg 4.37 (A) in CI; Python 3.11/3.12; CI green at 7b50775.
- ADR log 0001-0009 and CHANGELOG; router 68/68 on the dev set (blind 75% before patches, 20/20 after — that set is dev now, so a new hold-out is needed).
- Limitations: HNSW rebuilt on start (~60-70 s cold start, workaround for chroma#7090); single writer; no incremental indexing; gold set partly synthetic.

## LINKS AND CONTACT

**github.com/Smaturus/1c-mcp-rag-search · smaturus@gmail.com · MCP Python SDK · Chroma · multilingual-e5-small · BM25 · RRF · flashrank**

All numbers come from the repository evaluation reports (code_proc / gold-set v2, 4 runs).